Rakshak 2.0 - Round 3 Submission

Futurestack Hackathon - We Code Coders

Team & Project

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Project Name: Rakshak 2.0

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Live prototype: https://hariharan-007design.github.io/Rakshak2.0/

Repository: https://github.com/hariharan-007design/Rakshak2.0

What the product is

Rakshak 2.0 is a brand-governance review desk. A user pastes or uploads a draft, selects a company rule pack, runs a scan, and gets rule-cited findings with editable fixes. They accept or ignore each finding, then export a revised draft plus an audit report. Everything runs in the browser. No backend, no account, no cloud required for the deterministic scan.

Complete list of features built

- Select a demo company workspace (Mozilla, Mailchimp, Acme SaaS, Pharma MLR, Chemical GHS, FINRA) using a demo access code.

- Paste or upload .txt / .md copy, or a text-based PDF. PDF text extracted in-browser via bundled PDF.js.

- Compare draft against the active rule pack. Findings shown in order, highlighted in the draft. Each finding shows matched text, explanation, source link, editable suggestion.

- Apply or dismiss suggestions; review revised draft; copy or download a plain-text review report.

- Upload PNG/JPEG/WEBP/SVG logo for file-quality checks: dimensions, file size, transparency/clear-space heuristics. Labelled preliminary, not brand approval.

- Preview the supplied Mozilla Trademark Guidelines PDF and view the demo rule pack.

- Optional AI polish (BYO Groq/OpenAI-compatible key) rewrites a flagged sentence; key stays in browser; deterministic rules remain source of truth; output marked AI-generated and re-checkable.

- Channel preview for Website, LinkedIn, Instagram, X, Facebook (paragraph-break changes only).

- Saved files per industry stored in IndexedDB; saved-files view.

- Light/dark theme toggle.

- Sidebar maximize workspace, step indicator (Draft -> Review -> Approve).

- Drag-and-drop upload, empty-state handling, Analyze disabled when textarea empty.

Technical architecture and implementation

Layer: static web app (HTML/CSS/JS). No build step. Served over HTTP. Hosted on GitHub Pages.

Document handling: PDF.js (bundled, v6.4.299) extracts selectable text in-browser; explicit size/page limits surfaced.

Deterministic checks: regex against configured rule packs (JSON). High confidence for explicit rules; fast and predictable.

Findings/fixes: structured objects with rule ID + source URL. Side-by-side revised draft + downloadable report (diff-match-patch style).

Optional AI: OpenAI-compatible API (Groq), BYO key. Key in localStorage. Output marked AI-generated, human-reviewed.

Storage: localStorage (session only) + IndexedDB for saved files. No account, no server. Data not saved after tab closes for review history.

Original Round 1 target stack (Tauri v2 + Ollama + sqlite-vec) remains the post-hackathon desktop plan. Browser prototype chosen so judges can use it on any device with zero install.

Improvements completed after Round 2

Reliability under normal use:

- Fixed saved-files loading (drawFindings), industry filter, company name, PDF ref upload.
- Fixed drag-and-drop upload, step indicator.
- Disabled Analyze button when textarea empty.
- Removed navigator.onLine check blocking AI; clearer voice network error message.
- Dark mode CSS variables for workspace.
- Modal z-index, login form clickable, pointer-events fixes.

Usability:

- Sidebar maximize workspace, theme toggle in topbar, logo alignment, login redesign.
- Left-aligned logo, quote/login blur box, dot grid interaction.

Documentation:

- README with run instructions, demo workspaces, access codes, known limits, Round 3 roadmap.

The three improvements named at Round 2 - status:

1. Multimodal document & asset review (OCR, page-level evidence, approved-logo matching, accessibility checks): NOT delivered. Reason: a solo developer in 72 hours could not responsibly ship OCR, page-region citations, and approved-asset matching with tests. Declared as roadmap instead of half-working code. Partial progress: text-based PDF extraction already works; logo file-quality heuristics already work.

2. Enterprise governance & audit (real auth, RBAC, isolated workspaces, versioned rules, auditable approval history): NOT delivered as real security. Reason: the demo codes are explicitly documented as not authentication and must never protect real data. Shipping fake auth would mislead judges. The workspaces and decision history exist in-memory only.

3. Controlled publishing & quality measurement (approval-before-publishing, audit trails, FP/reviewer-override metrics): Partially delivered. Approval step indicator and in-memory decision history exist; publishing to platforms is not implemented (only channel preview with paragraph-break changes). Metrics are not tracked beyond the seeded test numbers in Round 2.

Honest note: Round 3 was spent finishing and hardening what Round 2 shipped (reliability, usability, docs) rather than starting the three large features. This matches the Round 3 brief's point 2 (close loose ends) and point 3 (hold up without us).

Testing details

Seeded test protocol (from Round 2, re-run after fixes):

- Violations detected (recall): 12/12 on seeded set (4 violations per pack, each mapped to one rule ID).
- False discovery: 0/20 clean control passages produced spurious findings.
- Deterministic scan speed: under 1 second on a representative draft.
- Offline operation: deterministic scan runs with no network; only optional AI polish calls an external API.

Manual end-to-end checks after fixes:

- Empty textarea -> Analyze disabled.
- Drag-and-drop upload of .txt/.md/.pdf.
- Refresh behaviour: review history is in-memory only (documented).
- Repeat action (Analyze twice) -> no crash, findings re-render.
- Bad input (empty, oversized PDF) -> surfaced limits, no hang.
- Theme toggle light/dark -> workspace readable in both.

Documentation

README.md covers: how to run (python -m http.server 8000 --directory web), live demo link, demo workspaces with access codes, what works, what it does not do, how to try, Round 3 roadmap, sources, repository layout. A judge can clone and run from the README.

Product roadmap and future development

- v0.2: VS Code / Figma / Chrome plugins sharing the agent core.
- v0.3: encrypted team vault (Syncthing/iCloud/Drive, no server).
- v1.0: optional SaaS dashboard for admins (opt-in).
- Community packs: Pharma MLR, Chem GHS/REACH, Fintech FINRA contributed back.
- Deliver the three Round-2-named improvements as real features (OCR + page citations; real auth/RBAC; publishing + metrics) in a follow-up release, not as half-working demo code.

Known limitations (honest boundaries)

- English only.
- Demo rule packs hand-curated; production PDF->JSON extraction is post-hackathon.
- No OCR; text-based PDFs only.
- Logo checks are file-quality heuristics, not brand approval.
- AI polish optional and BYO-key; output re-checkable and human-reviewed.
- Pre-screening aid only; not a compliance guarantee.